Systems and Network Administration

Assignment 1

Manual Execution

Student: Ahmed Mohamed Sabri Abusaliqa

Instructor: Dr. Jehad Hamamra

Date: October 12, 2026

1. `man ls`

```
LS(1)                                   General Commands Manual                                   LS(1)

NAME
       ls - List directory contents.  Ignore files and directories starting with a '.' by default

SYNOPSIS
       ls  [--help]  [--format]  [-C ] [-l|--long] [-x ] [-T|--tabsize] [-m ] [--zero] [-D|--dired] [--hyperlink] [-1 ]
       [-o  ]  [-g  ]  [-n|--numeric-uid-gid]  [--quoting-style]  [-N|--literal]  [-b|--escape]  [-Q|--quote-name]
       [-q|--hide-control-chars]  [--show-control-chars] [--time] [-c ] [-u ] [--hide] [-I|--ignore] [-B|--ignore-back-
       ups] [--sort] [-S ] [-t ] [-v ]  [-X  ]  [-U  ]  [-L|--dereference]  [--dereference-command-line-symlink-to-dir]
       [-H|--dereference-command-line]  [-G|--no-group]  [--author] [-a|--all] [-A|--almost-all] [-f ] [-d|--directory]
       [-h|--human-readable]  [-k|--kibibytes]  [--si]  [--block-size]  [-i|--inode]  [-r|--reverse]  [-R|--recursive]
       [-w|--width]  [-s|--size]  [--color]  [--indicator-style]  [-F|--classify]  [--file-type]  [-p  ] [--time-style]
       [--full-time] [-Z|--context] [--group-directories-first] [-V|--version] [paths]

DESCRIPTION
       List directory contents.  Ignore files and directories starting with a '.' by default

OPTIONS
       --help Print help information.

       --format
              Set the display format.

       -C     Display the files in columns.

       -l, --long
              Display detailed information.
```

2. `man 5 passwd`

```
PASSWD(5)                                File Formats and Configuration                                PASSWD(5)

NAME
       passwd - the password file

DESCRIPTION
       /etc/passwd contains one line for each user account, with seven fields delimited by colons (":"). These fields are:

       •   login name

       •   optional encrypted password

       •   numerical user ID

       •   numerical group ID

       •   user name or comment field

       •   user home directory

       •   optional user command interpreter

       If the password field is a lower-case "x", then the encrypted password is actually stored in the shadow(5) file instead; there must be a
       corresponding line in the /etc/shadow file, or else the user account is invalid.

       The encrypted password field may be empty, in which case no password is required to authenticate as the specified login name. However, some
       applications which read the /etc/passwd file may decide not to permit any access at all if the password field is blank.

       A password field which starts with an exclamation mark means that the password is locked. The remaining characters on the line represent the
       password field before the password was locked.

       Refer to crypt(3) for details on how this string is interpreted.

       If the password field contains some string that is not a valid result of crypt(3), for instance ! or *, the user will not be able to use a
 Manual page passwd(5) line 1 (press h for help or q to quit)
```

3. `man -k copy`

```
ahmad@AhmadSliqa:~$ man -k copy
Clone (3pm)          - recursively copy Perl datatypes
cp (1)               - copy files and directories
dd (1)               - Copy, and optionally convert, a file system resource
debconf-copydb (1)   - copy a debconf database
git-checkout-index (1) - Copy files from the index to the working tree
gnucp (1)            - copy files and directories
gnudd (1)            - convert and copy a file
gnuinstall (1)       - copy files and set attributes
install (1)          - Copy SOURCE to DEST or multiple SOURCE(s) to the existing DIRECTORY, while setting permission modes and owner/group
objcopy (1)          - copy and translate object files
rsync (1)            - a fast, versatile, remote (and local) file-copying tool
rust-cp (1)          - Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.
rust-dd (1)          - Copy, and optionally convert, a file system resource
rust-install (1)     - Copy SOURCE to DEST or multiple SOURCE(s) to the existing DIRECTORY, while setting permission modes and owner/group
rust-tee (1)         - Copy standard input to each FILE, and also to standard output.
scp (1)              - OpenSSH secure file copy
ssh-copy-id (1)      - use locally available keys to authorise logins on a remote machine
string_copying (7)   - copying strings and character sequences
tee (1)              - Copy standard input to each FILE, and also to standard output.
x86_64-linux-gnu-objcopy (1) - copy and translate object files
```

4. man -K “recursive”

```
BASH(1)                                  General Commands Manual                                 BASH(1)

NAME
       bash - GNU Bourne-Again SHell

SYNOPSIS
       bash [options] [command_string | file]

COPYRIGHT
       Bash is Copyright (C) 1989-2025 by the Free Software Foundation, Inc.

DESCRIPTION
       Bash is a command language interpreter that executes commands read from the standard input, from a string, or from a file.  It is a reimple-
       mentation  and extension of the Bourne shell, the historical Unix command language interpreter.  Bash also incorporates useful features from
       the Korn and C shells (ksh and csh).

       POSIX is the name for a family of computing standards based on Unix.  Bash is intended to be a conformant implementation of  the  Shell  and
       Utilities portion of the IEEE POSIX specification (IEEE Standard 1003.1).  Bash POSIX mode (hereafter referred to as posix mode) changes the
       shell's behavior where its default operation differs from the standard to strictly conform to the standard.  See SEE ALSO below for a refer-
       ence to a document that details how posix mode affects bash's behavior.  Bash can be configured to be POSIX-conformant by default.

OPTIONS
       All  of  the  single-character  shell options documented in the description of the set builtin command, including -o, can be used as options
       when the shell is invoked.  In addition, bash interprets the following options when it is invoked:

       -c        If the -c option is present, then commands are read from the first non-option argument command_string.  If there are arguments af-
                 ter the command_string, the first argument is assigned to $0 and any remaining arguments are assigned to  the  positional  parame-
                 ters.  The assignment to $0 sets the name of the shell, which is used in warning and error messages.

       -i        If the -i option is present, the shell is interactive.

       -l        Make bash act as if it had been invoked as a login shell (see INVOCATION below).

       -r        If the -r option is present, the shell becomes restricted (see RESTRICTED SHELL below).
 Manual page bash(1) line 1 (press h for help or q to quit)
```

5. cp fruits.txt fruits_backup.txt

```
ahmad@AhmadSliqa:~/Assignment1$ cp fruits.txt fruits_backup.txt
ahmad@AhmadSliqa:~/Assignment1$ ls
README.txt  cities.txt     first_names.txt  fruits_backup.txt  notes.txt    reports     server.log    todo.txt
archive     employees.txt  fruits.txt       last_names.txt     numbers.txt  scores.csv  students.txt
```

6. cp cities.txt my_folder

```
ahmad@AhmadSliqa:~/Assignment1$ mkdir my_folder
ahmad@AhmadSliqa:~/Assignment1$ cp cities.txt my_folder
```

7. cp -r my_folder backup_folder

```
ahmad@AhmadSliqa:~/Assignment1$ cp -r my_folder backup_folder
ahmad@AhmadSliqa:~/Assignment1$ ls backup_folder/
cities.txt
```

8. mv first_names.txt names.txt (rename)

```
ahmad@AhmadSliqa:~/Assignment1$ mv first_names.txt names.txt
ahmad@AhmadSliqa:~/Assignment1$ ls
README.txt  backup_folder  employees.txt  fruits_backup.txt  my_folder  notes.txt    reports     server.log    todo.txt
archive     cities.txt     fruits.txt     last_names.txt     names.txt  numbers.txt  scores.csv  students.txt
```

9. mv notes.txt my_folder/ (move)

```
ahmad@AhmadSliqa:~/Assignment1$ mv notes.txt my_folder/
ahmad@AhmadSliqa:~/Assignment1$ ls my_folder/
cities.txt  notes.txt
```

10. `cat notes.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ cat fruits.txt
orange
apple
banana
apple
grape
banana
kiwi
orange
apple
mango
```

11. `more notes.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ more notes.txt
Linux is a multiuser operating system.
The terminal accepts commands.
A shell interprets commands and starts programs.
Files can be copied, moved, renamed, and removed.
Directories organize files.
The grep command searches text.
The sort command sorts lines.
Ubuntu can run on Windows using WSL.
Practice makes command-line work easier.
```

12. `less notes.txt`

```
Linux is a multiuser operating system.
The terminal accepts commands.
A shell interprets commands and starts programs.
Files can be copied, moved, renamed, and removed
Directories organize files.
The grep command searches text.
The sort command sorts lines.
Ubuntu can run on Windows using WSL.
Practice makes command-line work easier.
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
(END)
```

13. `rm fruits.txt cities.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ rm fruits.txt cities.txt
ahmad@AhmadSliqa:~/Assignment1$
```

14. `rmdir my_folder`

```
ahmad@AhmadSliqa:~/Assignment1$ rmdir my_folder
ahmad@AhmadSliqa:~/Assignment1$
```

15. `rm -r my_folder`

```
ahmad@AhmadSliqa:~/Assignment1$ ls my_folder/
1.txt  2.txt
ahmad@AhmadSliqa:~/Assignment1$ rm -r my_folder
ahmad@AhmadSliqa:~/Assignment1$
```

16. `ls -la`

```
ahmad@AhmadSliqa:~/Assignment1$ ls -al
total 64
drwxr-xr-x 5 ahmad ahmad 4096 Oct 12 19:06 .
drwxr-x--- 8 ahmad ahmad 4096 Oct 12 19:01 ..
-rw-r--r-- 1 ahmad ahmad  396 Oct 11 20:36 README.txt
drwxr-xr-x 2 ahmad ahmad 4096 Oct 11 23:38 archive
drwxr-xr-x 2 ahmad ahmad 4096 Oct 12 18:52 backup_folder
-rw-r--r-- 1 ahmad ahmad  116 Oct 11 20:36 employees.txt
-rw-r--r-- 1 ahmad ahmad   63 Oct 12 18:18 fruits_backup.txt
-rw-r--r-- 1 ahmad ahmad   40 Oct 11 20:36 last_names.txt
-rw-r--r-- 1 ahmad ahmad   35 Oct 11 20:36 names.txt
-rw-r--r-- 1 ahmad ahmad  337 Oct 11 20:36 notes.txt
-rw-r--r-- 1 ahmad ahmad   29 Oct 11 20:36 numbers.txt
drwxr-xr-x 2 ahmad ahmad 4096 Oct 11 23:38 reports
-rw-r--r-- 1 ahmad ahmad  195 Oct 11 20:36 scores.csv
-rw-r--r-- 1 ahmad ahmad  464 Oct 11 20:36 server.log
-rw-r--r-- 1 ahmad ahmad   46 Oct 11 20:36 students.txt
-rw-r--r-- 1 ahmad ahmad  104 Oct 11 20:36 todo.txt
```

17. `mkdir practice_folder`

```
ahmad@AhmadSliqa:~/Assignment1$ mkdir practice_folder
ahmad@AhmadSliqa:~/Assignment1$ ls
README.txt  backup_folder  fruits_backup.txt  names.txt  numbers.txt      reports     server.log  todo.txt
archive     employees.txt  last_names.txt     notes.txt  practice_folder  scores.csv  students.txt
```

18. `cd practice_folder`

```
ahmad@AhmadSliqa:~/Assignment1$ cd practice_folder/
ahmad@AhmadSliqa:~/Assignment1/practice_folder$
```

19. `pwd`

```
ahmad@AhmadSliqa:~/Assignment1/practice_folder$ pwd
/home/ahmad/Assignment1/practice_folder
```

20. `head -n 3 employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ head -n 3 employees.txt
E101:Amal:IT:4200
E102:Khaled:HR:3500
E103:Rana:IT:4600
```

21. `tail -n 3 employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ tail -n 3 employees.txt
E104:Basel:Sales:3900
E105:Salma:HR:3700
E106:Tariq:IT:4400
```

22. `wc -l employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ wc -l employees.txt
6 employees.txt
```

23. `wc -w fruits_backup.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ wc -w fruits_backup.txt
10 fruits_backup.txt
```

24. `wc -c fruits_backup.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ wc -c fruits_backup.txt
63 fruits_backup.txt
```

25. `sort fruits_backup.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ sort fruits_backup.txt
apple
apple
apple
banana
banana
grape
kiwi
mango
orange
```

26. `sort -r fruits_backup.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ sort -r fruits_backup.txt
orange
orange
mango
kiwi
grape
banana
banana
apple
apple
```

27. `sort -t: -k 1 employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ sort -t: -k 1 employees.txt
E101:Amal:IT:4200
E102:Khaled:HR:3500
E103:Rana:IT:4600
E104:Basel:Sales:3900
E105:Salma:HR:3700
E106:Tariq:IT:4400
```

28. `sort -t: -n -k 4 employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ sort -t: -n -k 4 employees.txt
E102:Khaled:HR:3500
E105:Salma:HR:3700
E104:Basel:Sales:3900
E101:Amal:IT:4200
E106:Tariq:IT:4400
E103:Rana:IT:4600
```

29. `cut -f2 -d: employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ cut -f2 -d: employees.txt
Amal
Khaled
Rana
Basel
Salma
Tariq
```

30. `grep “IT” employees.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ grep "IT" employees.txt
E101:Amal:IT:4200
E103:Rana:IT:4600
E106:Tariq:IT:4400
```

31. `paste first_names.txt last_names.txt`

```
ahmad@AhmadSliqa:~/Assignment1$ paste first_names.txt last_names.txt
Omar    Hassan
Abd     Nasser
Yaseen  Saleh
Ahamd   Khatib
Odeh    Odeh
```

32. `tr ‘a’ ‘o’ < notes.txt (replace a to o)`

```
ahmad@AhmadSliqa:~/Assignment1$ tr 'a' 'o' < notes.txt
Linux is o multiuser operoting system.
The terminol occepts commonds.
A shell interprets commonds ond storts progroms.
Files con be copied, moved, renomed, ond removed.
Directories orgonize files.
The grep commond seorches text.
The sort commond sorts lines.
Ubuntu con run on Windows using WSL.
Proctice mokes commond-line work eosier.
```